Aman Verma

AI/ML ENGINEERING INTERN

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EXPERIENCE

Software Development Engineer Intern January 2026 - June 2026

LexoLegal *Remote*

- Architected asynchronous REST APIs in **FastAPI**, **SQLAlchemy**, and **PostgreSQL** (SQL) behind an **Nginx** proxy on **Linux**, containerized with **Docker** with automated **Git** and **GitHub Actions** CI/CD pipelines.
- Built and deployed 'Munshi', an autonomous legal assistant using **Pydantic AI** and **FastAPI**, resolving multi-turn case inquiries over WhatsApp with **sub-2s latency** via tool-calling execution loops.
- Prevented race conditions and dropped turns from rapid message bursts by designing **Celery** workers with atomic **Redis** distributed locks and non-blocking 10s countdown task requeues (90-retry ceiling).
- Integrated **Model Context Protocol (MCP)** clients with **Pydantic AI** to interface external tool servers, isolating verified user identity inside AgentDeps rather than LLM tool parameters to eliminate prompt injection risks.
- Automated daily judicial data ingestion across all 25 High Courts and 743 District Courts via **Celery** and **Bash** scripts on **Linux**, utilizing **Selectolax** DOM extractors and batch **SQL** inserts into **PostgreSQL** to keep records synchronized.

PROJECTS

ncert-rag-evals | *Python 3.14, uv, Ruff, SQLite FTS5, ChromaDB, Pytest* GitHub

- Constructed an empirical retrieval benchmark across 17 STEM textbooks, implementing custom hybrid search from scratch in **SQLite FTS5** and **ChromaDB** to profile 8 retrieval strategies on raw CPU latency.
- Proved that fusing BM25 with dense embeddings via Reciprocal Rank Fusion (RRF) achieved **86.9% Recall@5 in 8.9ms on CPU**, matching LLM query expansion while cutting latency by **99.8%**.
- Engineered a document parser indexing 4,429 section-aware chunks into **SQLite FTS5** and **ChromaDB**; validated statistical significance via paired McNemar tests (**p = 0.0021**) across 92 unit tests in **Pytest** with **Ruff** linting.

doubtless | *Pydantic AI, FastAPI, Multimodal RAG, Groq Whisper, Celery, Redis, Logfire, OpenTelemetry* GitHub

- Engineered an interactive lecture doubt-solver in **Pydantic AI**, **Asyncio**, and **FastAPI**, fusing blackboard video frames with 45s dialogue windows to stream real-time solutions for visual STEM queries via SSE.
- Designed an asynchronous 4-stage media ingestion pipeline (PipelineRunner) in **Celery** and **Redis** that transcodes HLS streams, slices 16kHz audio (<15MB/hr), and transcribes in <10s via **Groq Whisper-Large-V3**.
- Architected the async data layer with thread-safe **SQLite WAL** repositories and **Redis** caching; instrumented distributed tracing across agent runs via **OpenTelemetry** and **Logfire** (151 tests).

vectorless-rag | *Python, Pydantic AI, Ollama, ChromaDB, PyMuPDF* GitHub

- Evaluated dense vector search against LLM-based page indexing on an identical document corpus using **Pydantic AI**, configuring **Ollama** as a local testbed alongside cloud LLMs across 18 benchmark queries.
- Benchmarked retrieval trade-offs: generative page indexing achieved **100% retrieval hit rate** and **0.85 citation precision** (vs 93% and 0.69 for vector RAG), while quantifying the 2.7x token cost and 2x latency tradeoff.
- Implemented hand-written ingestion, page-offset tracking, and request batching from scratch in **Python** using **PyMuPDF** and **ChromaDB**, deliberately bypassing high-level framework wrappers to isolate raw retrieval diffs.

TECHNICAL SKILLS

Languages & Systems: Python, SQL, Bash, Linux

AI & Agent Frameworks: Pydantic AI, Model Context Protocol (MCP), Ollama, ChromaDB, Groq Whisper

Backend & Distributed Infrastructure: FastAPI, Celery, Redis, PostgreSQL, SQLite, Nginx, Asyncio

Evals, Benchmarking & Testing: Pytest, OpenTelemetry, Logfire, PyMuPDF, Selectolax

Developer Tools & DevOps: Docker, Git, GitHub Actions, uv, Ruff

LEADERSHIP & ACTIVITIES

- Delivered 3 technical sessions as a speaker on 'Generative AI Unlocked: LLM Basics' and 'Level Up Your Career with AI', training 150+ college students on foundational LLM concepts and modern AI developer tools.
- Organized Smart India Hackathon internal rounds across 2 editions; coordinated problem statement curation, mentoring, and judging logistics for 20+ teams and 120+ participants.

EDUCATION

Veer Bahadur Singh Purvanchal University 2023 - 2027

Bachelor of Technology in Computer Science & Engineering *Jaunpur, India*